Eclipses by Appointment: 104 Uncatalogued Eclipsing Systems, Catalogue Corrections and a Measured Null Planet Search from Public TESS and Gaia DR3 Data

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The author's role and the use of AI systems are set out in Section 6, Contributions and AI use. The per-object entries for the eclipsing systems, one for each with its light curve, are in the Supplement; the entries for the catalogue corrections, the probable blends A13–A15 and the other objects are in the submitted version of the paper (Appendix B).

AAVSO keywords = Eclipsing Binaries; Catalogues, Databases, Surveys; Period Analysis, Period Change; Extrasolar Planets; Variable Stars (individual), Observing Targets: HIP 1060, HD 170829, V1192 Cas, TYC 4590-364-1

ADS keywords = binaries: eclipsing; binaries: spectroscopic; catalogs; planetary systems; stars: variables: general; techniques: photometric

ABSTRACT

We report 104 eclipsing systems found in public TESS photometry whose eclipses are in no vetted variable-star or eclipsing-binary catalogue: 93 confirmed and 11 probable. Fifty-four come from a method we call eclipses by appointment: a Gaia DR3 spectroscopic orbit with a period of 20–60 d predicts when its conjunctions fall, and the TESS light curve is searched only inside those windows, where a blind period search is far less effective. Among 3,749 stars with G < 11 the method found about 1.4 new eclipsing binaries per hundred stars. Forty-six more came from period folds of Gaia orbits shorter than 20 d and from transit searches of nearby M dwarfs, and four from edge-on Gaia astrometric orbits. Every candidate from the appointments search, the G 9–10 fold and the edge-on search was re-analysed independently with separately written code. The eclipses contradict twelve Gaia DR3 orbit solutions in period or $e \cos \omega$; six eccentric binaries are new heartbeat stars and a seventh is a candidate; eleven existing catalogue entries are corrected. For 17 of 31 deep systems examined, Harvard plates from 1890–1990 show the eclipse at the ephemerides given here, ASAS-SN photometry from 2014–2018 shows it for 20 of 55 systems tested, with measured depths matching the predicted ones in the 11 examined, and ASAS-3 photometry from 2000–2009 shows it for 10 of 34. A transit search of about 3,100 nearby M dwarfs, from which the project began, found no new planet where about 1.2 detections were expected under the estimator's assumptions; we recovered both TIC 4206066 candidates and show TOI-5394.01 to be the occultation of an M-dwarf companion. Ephemerides and next-eclipse predictions are given for every confirmed and probable system but one, so that observers can time the eclipses.

1. Introduction

Gaia DR3 published about 180,000 spectroscopic orbits (Gaia Collaboration, Arenou et al. 2023), each giving a period, an eccentricity, an argument of periastron and a time of periastron for a binary resolved only in velocity. TESS has photometered most of those stars in its full-frame images over the same years. Where the two overlap, an orbit seen within a degree or two of edge-on must eclipse, and TESS records the eclipse whenever a conjunction falls in an observed sector. For periods below about 20 d a period search at Gaia's period finds such eclipses easily. Between 20 and 60 d it does not: a 27-day sector holds at most one eclipse of each kind, and multi-sector baselines of years can be phased only when the period is already known to minutes or the single events are deep enough to be found one by one. Catalogued eclipsing binaries are correspondingly scarce in that period range, and the Gaia orbits that would gain most from an eclipse, which fixes the inclination and with it the companion mass, are the ones TESS searches least well.

Gaia DR3 contains about 800,000 non-single-star solutions for astrometric, spectroscopic and eclipsing binaries (Gaia Collaboration, Arenou et al. 2023). The spectroscopic processing, restricted to single-lined F–K stars brighter than magnitude 12, delivered 181,327 SB1 orbits after filtering (Gosset et al. 2025), and the astrometric processing about 165,500 orbits (Halbwachs et al. 2023a). The SB1 orbits are mostly reliable: of 827 binaries shared with the SB9 catalogue, 655 agree to within 10% in period and eccentricity (Merle et al. 2026), and a comparison with LAMOST and GALAH velocities removed the physically unlikely short-period, high-eccentricity solutions (Bashi et al. 2022). Double-lined solutions fare worse, matching the photometry in period and eccentricity for about half of SB2 eclipsing binaries (Rowan et al. 2023) and 20% of SB2 heartbeat stars, against 85% of SB1 ones (Callahan et al. 2025).

TESS eclipsing binaries are catalogued by Prša et al. (2022; 4,584 systems from sectors 1–26), IJspeert et al. (2021, 2024) and Kostov et al. (2025; 10,001 vetted systems from the full-frame images and about 900,000 unvetted machine-learning detections), and Gaia DR3 adds 2,184,477 candidates from its own photometry (Mowlavi et al. 2023). All favour short periods: the 7,936 new systems of Kostov et al. (2025) have a median period of 3.5 d and a 95th percentile of about 11.6 d. Prša et al. (2022) attribute this to the sector length, which suppresses detections from periods as short as 13 d, and to the box least-squares search, which cannot find single eclipses. Planet searches meet the same limit with single-event methods, estimating the period from one transit's shape (Osborn et al. 2016) or catching a second event

from the ground, which also finds eclipsing binaries such as a 61.8-d M dwarf eclipsing a Sun-like star (Gill et al. 2020); "duotransits", one transit in each of two TESS years, typically have periods above 20 d (Hawthorn et al. 2024). In each case the event is found first and the period afterwards.

Heartbeat stars are eccentric binaries that brighten near periastron through tidal distortion, irradiation and Doppler beaming (Kumar et al. 1995; Welsh et al. 2011; Thompson et al. 2012); some show tidally excited oscillations (Fuller and Lai 2012; Fuller 2017). Kepler supplied the first samples (Kirk et al. 2016; Shporer et al. 2016), and the TESS searches have been at short periods: 23 systems at P = 2.7–20 d (Li et al. 2024a), 180 at $P \lesssim 10$ d (Solanki et al. 2025) and 112 from Gaia spectroscopic orbits, with the SB1 orbits cut at P < 13 d (Callahan et al. 2025). A third body shows in eclipse times through the light-travel-time effect (Irwin 1952); eclipse timing found third bodies in 222 of about 2,600 Kepler binaries (Rappaport et al. 2013; Borkovits et al. 2016), and 403 known eclipsing binaries have a Gaia DR3 solution at a period much longer than the eclipse period (Czavalinga et al. 2023a). Over longer baselines, the Harvard plates, about 500,000 taken from 1885 to 1992 and digitised by DASCH with photometry good to about 0.1–0.15 mag (Laycock et al. 2010; Grindlay et al. 2012; Tang et al. 2013; Williams 2025), have been combined with TESS eclipse times to follow a binary's period over a century (Zhang et al. 2026).

New variables are registered in the AAVSO International Variable Star Index (VSX; Watson et al. 2006); times of minimum from AAVSO observers appear regularly in this journal (e.g., Samolyk 2022), and eclipsing binaries found in survey and archival data have been reported here, entered in VSX and characterised with the authors' own photometry (Romanov 2021; De Elías 2025). Guinan et al. (2012) set out the follow-up that eclipsing binaries from space missions need from observers.

Orbits and eclipses have been combined in both directions. The EBLM programme measured orbits for 118 binaries first seen as shallow WASP eclipses (Triaud et al. 2017), and joint fits of this kind supply the masses and radii in DEBCat (Southworth 2015). Starting from Gaia, Rowan et al. (2023) measured masses and radii of 122 stars from SB2 orbits and ASAS-SN light curves, Schmidt et al. (2023) matched SB1 orbits to TESS Objects of Interest, and Halbwachs et al. (2023b) selected 41 bright astrometric binaries that could eclipse. The conjunction prediction itself is not new: Tarrants and Mendes (2023) applied it to Gaia DR3 SB1 elements to vet known TESS and K2 planet candidates and called it their least reliable test, and Yamaguchi et al. (2024) applied it to one 358-d orbit in search of self-lensing pulses. Eclipses in Gaia orbits have been found by other routes: Callahan et al. (2025) found 18 among their 112 heartbeat candidates with a periodogram, El-Badry et al. (2026) found 25 among the orbits of compact-object candidates, 14 of them hierarchical triples with an eclipse period much shorter than Gaia's, and Beck et al. (2024) searched the six oscillating red giants whose Gaia astrometric inclinations are within 3° of edge-on, a sample that includes our A60, which they did not flag. We are not aware of a previous search that uses the conjunction times predicted by Gaia DR3 spectroscopic orbits as the search itself, examining TESS only inside those windows at P = 20–60 d, where catalogued TESS eclipsing binaries are scarce. That statement rests on a web and arXiv search; the step is the one taken here.

The work reported here began from a planet question. A public report of two transiting-planet candidates around the K dwarf TIC 4206066 (Rabtsevich 2026) led to a transit search of about 3,100 nearby M dwarfs in TESS full-frame-image light curves (Section 2.3). That search found no new planet (Section 4.6), recovered both reported candidates without using the reported periods or epochs, and turned up four uncatalogued eclipsing systems (A1–A4), two of which had Gaia DR3 evidence of a stellar companion that had never been combined with the TESS photometry. We therefore inverted the search: starting from Gaia's spectroscopic orbits, we tested each for eclipses in TESS. For orbits shorter than 20 d this is a period search at Gaia's period (Section 2.4), and it found 42 uncatalogued eclipsing systems.

For longer periods a Gaia orbit still fixes the conjunction times: the period, the time of periastron, the eccentricity and the argument of periastron together determine when each conjunction falls, to about 2 d at TESS epochs for the median orbit in our sample. Eclipses can only happen at conjunction, so the search can be confined to the predicted windows and use a single-event statistic instead of a period search over years of data. The number of trials falls from a four-year period grid to the windows, which hold about half of the usable data for the median star, and an eclipse that lands on the predicted conjunction is attributed to the star with the Gaia orbit, which is the hardest step in any blended-field TESS search. We call this eclipses by appointment (Section 2.5). It produced 54 of the 104 systems in this paper.

What is new here is (i) the appointments method and its yield, about 1.4 new eclipsing binaries per hundred Gaia orbits searched at G < 11; (ii) the 104 systems, with ephemerides and next-eclipse predictions (Table A1); (iii) eleven catalogue corrections and twelve Gaia DR3 orbit solutions that the eclipses contradict; and (iv) a null planet result with its expected yield estimated under stated assumptions, and the reason it was null. "Uncatalogued" means absent from the vetted catalogues and the literature named in Section 2.7, not unseen by any pipeline: 72 of the 104 carry an automated threshold-crossing event of the SPOC or TESS-SPOC pipelines (Jenkins et al. 2016; Caldwell et al. 2020) that was never vetted or promoted to a TESS Object of Interest, and a full-text search of NASA ADS found a prior mention of eclipses for only one (A7; Bryant et al. 2023). The object labels (A1–A109, B1–B11, C1–C3) were assigned in this work and are used as identifiers throughout; the per-object entries, one for each system with its light curve, are in the Supplement, and the data and code are in the deposit described in Appendix B.

2. Data and methods

2.1 TESS photometry

All photometry is from TESS (Ricker et al. 2015), retrieved from MAST. The primary source was the Quick-Look Pipeline (QLP; Huang et al. 2020; Kunimoto et al. 2021) full-frame-image light curves, at 30-min (sectors 1–26), 10-min (27–55) and 200-s (56 onward) cadence. For the long-period work we used the raw aperture flux SAP_FLUX (Section 2.8). Independent reductions of the same pixels served as cross-checks: SPOC 2-min (Jenkins et al. 2016), TESS-SPOC (Caldwell et al. 2020), TGLC (Han and Brandt 2023), eleanor (Feinstein et al. 2019), GSFC-ELEANOR-LITE (Powell et al. 2022), TASOC (Handberg et al. 2021), CDIPS (Bouma et al. 2019), TARS (Boyle et al. 2026), T16 (Hartman et al. 2025) and TEQUILA (Ogunwale et al. 2025). Times are BTJD = BJD_{TDB} − 2457000.

2.2 Gaia DR3 orbits

Orbits are from the Gaia DR3 (Gaia Collaboration, Vallenari et al. 2023) table nss_two_body_orbit (Gaia Collaboration, Arenou et al. 2023; Halbwachs et al. 2023a; Gosset et al. 2025): the spectroscopic types SB1, SB2, SB1C, SB2C and AstroSpectroSB1, and the astrometric Orbital type (Section 2.6), matched to the TESS Input Catalog (TIC 8.2; Stassun et al. 2019; Paegert et al. 2021) by Gaia source identifier.

Gaia gives the time of periastron T_p relative to J2016.0 = BTJD 389.0. Conjunction A is at true anomaly ν = 90° − ω and conjunction B at ν = 270° − ω, with ω the argument of periastron of the star Gaia tracks, which is behind its companion at A. From ν follow the eccentric anomaly E, the mean anomaly M = E − e sin E, and

$$t_c = T_p + 389.0 + M\,P / 2\pi \ \ (\mathrm{BTJD}),\ \ \textit{plus integer multiples of } P.$$

Uncertainties were propagated by Monte Carlo through Gaia's errors on P, T_p, e and ω; the conjunction tests used Gaia's full covariance matrix. For circular solutions ω = 0 was assumed. “Primary” denotes the deeper eclipse; a transit is the smaller star crossing the larger, an occultation the smaller star hidden.

2.3 Transit searches of nearby M dwarfs

The parent sample was the TIC 8.2 dwarfs with T_{eff} 3200–4300 K, Tmag 8.5–11 and distance < 40 pc, less every star with a TESS Object of Interest (TOI) or community TOI and every star with a TIC contamination ratio of 0.05 or more; 1,299 of these had 3–8 QLP sectors. Further searches covered fainter and smaller stars, crowded fields, single transits, periods of 0.17–0.5 d and 78 known planet hosts. The searches used box least-squares (BLS; Kovács et al. 2002), with completeness measured from injected transits. Surviving peaks faced odd/even tests, a cyclic-roll null that keeps red noise and destroys transits, difference-image centroids in two field orientations, and archive queries.

2.4 Period folds of Gaia orbits with P < 20 d

The first sample was every Gaia DR3 spectroscopic orbit with P < 20 d and parallax > 10 mas: 736 science orbits, plus 14 Gaia eclipsing binaries as controls, all recovered. A coherent BLS over the flattened QLP flux searched ±max($5\sigma_P$, 0.1%) around P_{Gaia} and $P_{Gaia}/2$, scoring each star with a z value against 300 random periods on the same star. Triage removed VSX matches, rotators and ellipsoidal variables, and required an eccentric system's eclipse to land on Gaia's predicted conjunction.

The same code then ran on orbits with parallax ≤ 10 mas or missing. At G < 9, an independent re-check kept seven (A8–A12, B6, B7). At 9 ≤ G < 10, 2,559 science targets among 3,022 orbits were searched, and 199 of 218 controls (91%) reached z ≥ 5, which measures the search's completeness. After triage and catalogue searches, 38 folds showed eclipse-shaped dips and 37 were re-analysed to the full protocol of Section 2.7. About 175 eclipse-like stars with a VSX entry within 30″ and nine marginal dips were not pursued, so this yield is a lower bound.

2.5 Eclipses by appointment

Sample and windows. Gaia DR3 holds 11,446 spectroscopic orbits with 20 ≤ P < 60 d and G < 12. We searched two subsamples, G < 10 (1,422 stars, including 40 Gaia eclipsing binaries as controls) and G 10–11 (2,327 stars): 3,749 stars in all. For each orbit the conjunction times and their uncertainty σ_T were computed at every TESS cycle (Section 2.2). Each window had a half-width of max($3\sigma_T$, 0.25 d), capped at 7 d.

Statistic and cuts. In each window the QLP SAP_FLUX (15-min bins) was searched with a single-event box of 1–14 h against the mean of two side windows 0.75–2.5 durations out. Two shape tests followed: the far ratio (the depth against a baseline 3–6 durations out, divided by the near depth, about 1 for an eclipse) and a deficit in both halves of the box. The null was the same statistic in 40 random windows on the same star outside every predicted window, giving z = (SNR − median)/MAD, a rank against the star's own null rather than a Gaussian significance. A candidate needed SNR ≥ 8, a far ratio in [0.6, 1.6], the event within $3\sigma_T$ of a conjunction, no null window reaching it, and z ≥ 8 where a null exists. Of the 40 controls, 32 showed an eclipse at the predicted conjunction.

Coverage. Over the 3,708 science stars the windows hold a median 51% of the usable cadences, and every cadence for 165 stars (4.5%), where no null exists; for 591 more (16%), z is undefined because all null windows land on the same feature (MAD = 0). Seven of the 54 reported systems therefore never

faced the z test (A37, A40 and A42 with no null; A27, A51, A54 and A64 with MAD = 0) and passed on the remaining cuts. One, A18, failed a cut (far ratio 2.19); it was taken from the raw SNR ranking before the cuts were written and rests on six eclipses and its re-analysis.

Classification. Candidates passing the cuts were matched against VSX, ASAS-SN, Prša et al. (2022), IJspeert et al. (2024), Kostov et al. (2025) and SIMBAD; a VSX entry of a non-eclipsing type did not remove a star. The remaining 161 (G < 10) and 280 (G 10–11) stars were classified visually. Every star classified as an eclipser, and every unusual non-eclipse signal, was re-analysed independently (Section 2.7), with three exceptions (Section 3.1): 16 at G < 10 and 54 at G 10–11.

2.6 Edge-on Gaia astrometric orbits

Of the 6,652 Gaia DR3 astrometric orbits with P < 300 d and G < 13, we kept the 191 whose inclination, recovered from the Thiele–Innes coefficients, is within 2° of edge-on; the inclination served only to rank. For the 177 with QLP light curves, raw SAP flux was searched around Gaia's period and its half, and separately with a per-sector box search over 0.3–13 d and an inspection of the deepest single events.

2.7 Independent re-analysis and catalogue checks

Re-analysis. Every candidate from the appointments search, the G 9–10 fold and the edge-on search was re-analysed independently, with separately written code and with the aim of refuting the claim (Section 6). Each re-analysis re-measured every eclipse from raw SAP flux, including flagged cadences; fitted a linear ephemeris to all events, so that the period comes from the eclipses themselves; located the flux deficit in difference images relative to the target and every Gaia neighbour bright enough to produce it; re-measured the eclipses in at least one independent pipeline; compared the eclipse geometry with Gaia's orbit and mass function; and predicted the next eclipses.

Catalogues. Each object was cone-searched in VSX (Watson et al. 2006), SIMBAD (Wenger et al. 2000), the Gaia DR3 eclipsing-binary (Mowlavi et al. 2023), classifier (Rimoldini et al. 2023; Eyer et al. 2023) and non-single-star tables, Prša et al. (2022), IJspeert et al. (2024), Kostov et al. (2025) and ASAS-SN (Shappee et al. 2014; Kochanek et al. 2017; Jayasinghe et al. 2018), through VizieR (Ochsenbein et al. 2000), with a known eclipsing binary as positive control. A repeat on 2026 October 13, adding the GCVS and NSV (Samus' et al. 2017), ASAS-3 (Pojmański 2002), ATLAS (Heinze et al. 2018), ZTF (Chen et al. 2020) and Catalina (Drake et al. 2014), found no eclipsing-type entry for any counted A-object. A full-text query of the NASA ADS index for 248 identifiers of the 74 A-objects (2026 October 12) returned one prior report of eclipses (A7; Bryant et al. 2023); a second (October 13) covered 242 identifiers of the 35 G 9–10 systems.

Status words. **Uncatalogued** means that the eclipses were absent, at the time of the check (2026 October 11–12), from VSX, the SIMBAD variability types, the Gaia DR3 eclipsing-binary solutions, Prša et al. (2022), IJspeert et al. (2024), the vetted tables of Kostov et al. (2025) and ASAS-SN; an entry only in the unvetted machine-learning list of Kostov et al. (2025, table 2) leaves an object uncatalogued. **Confirmed**: the system passed every check above that its data allow (repeated events in raw flux, a period from the eclipses, a difference image on the target, a second pipeline, a geometry consistent with Gaia's orbit). **Probable**: the eclipses are real and repeat, but one check is missing or failed. **Plausible**: a single event, or two with the period untested. **By-catch**: a real variable on the target whose signal does not follow the Gaia orbit that selected it.

2.8 Known systematics

QLP's detrended flux suppresses long eclipses in post-sector-56 files: it removed the 20-h primary of A16. QLP's sector-80 light curves run 2.7 min late against SPOC and TESS-SPOC in the two systems where this was measured (A11, A98); whether the offset applies to the whole sector has not been tested. At 30-min cadence QLP mid-times can be off by minutes: in A92, sectors 14 and 15 are about 6 min late against three independent pipelines. Depths are those measured in the named pipeline and are diluted by any neighbour it does not correct for. Because the QLP aperture changes between sectors, a neighbour can make one sector's eclipse 9–27% shallower than the others (for example A85), so depths are taken from the sectors that agree with difference images or a second pipeline.

3. Results

3.1 Sample and yield of the appointments search

Table 1 gives the yield of the appointments search stage by stage. In the G < 10 subsample, 181 stars passed the cuts. In the G 10–11 subsample, 447 of 2,327 stars passed, and the 280 that no eclipsing-binary catalogue lists were inspected. Of the 70 stars re-analysed independently, 54 are eclipsing systems: 50 confirmed and four probable (A37, A40, A53, A68). Three are new heartbeat stars (Section 3.3). Table 2 lists the rest. Four candidates were refuted, three remain plausible, three on-target stars were already known, and 20 catalogued eclipsing binaries were recovered blind in each subsample.

The automated search missed whole eclipses in two ways. The search first discarded 0.5 d after every data gap, which removed whole eclipses (five of the six of A18); the trim was reduced to 0.1 d and the G < 10 subsample was searched again. And in many systems whole eclipses lie in cadences that QLP flags, mostly for scattered light, although the flux is usable: five of the ten eclipses of A30, for example. These were recovered from flagged cadences, other pipelines or cutout photometry.

3.2 The eclipsing systems

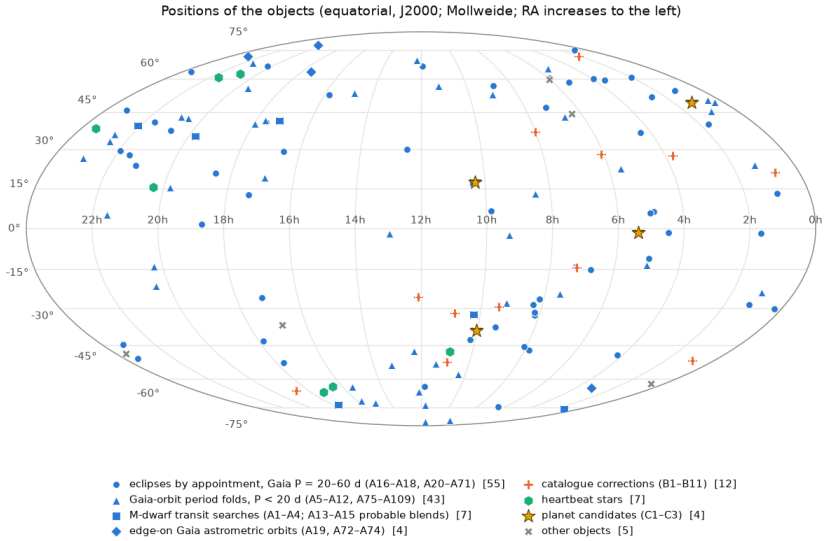


Figure 1. Positions of the objects (equatorial, J2000, Mollweide projection). Marker shape gives the search that found each eclipsing system; the catalogue corrections, heartbeat stars, planet candidates and other objects are shown separately. Counts are in brackets.

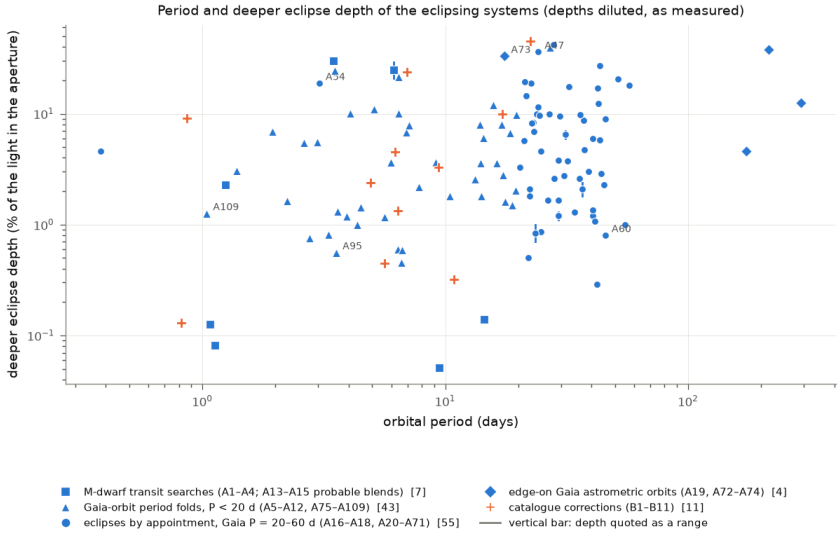


Figure 2. Orbital period and deeper eclipse depth of the eclipsing systems and the catalogue corrections. Depths are as measured in the pipeline aperture, diluted by any blended light; a vertical bar spans a depth quoted as a range. The appointments search (circles) fills the 20–60 d range that the period folds (triangles) do not reach.

Table A1 lists the 104 systems, 93 confirmed and 11 probable, with ephemerides and next eclipses; Figure 1 shows their positions and Figure 2 their periods and depths. Periods run from 1.0394878 d (A109) to 290.9037 d (A19).

HIP 1060 (A47) illustrates the method (Figure 3). It shows a 34%, 6.9-h total eclipse at conjunction A of its Gaia double-lined orbit and a 36%, 16-h annular transit at B, eleven events over eight years at P = 24.10512 d. Because Gaia measured both velocities, one model gives both masses (1.10 and 1.22 $M_{\odot}$) and radii (1.26 and 1.83 $R_{\odot}$). Three eclipses in sector 106, predicted before those data were examined, fell within 1.5 min of the ephemeris. HD 170829 (A20), at G 6.3 the brightest star here, is a G8 subgiant with an asteroseismic mass, eclipsed every 26.384493 d; VSX lists it as NSV 24480 with type “VAR:” and no period.

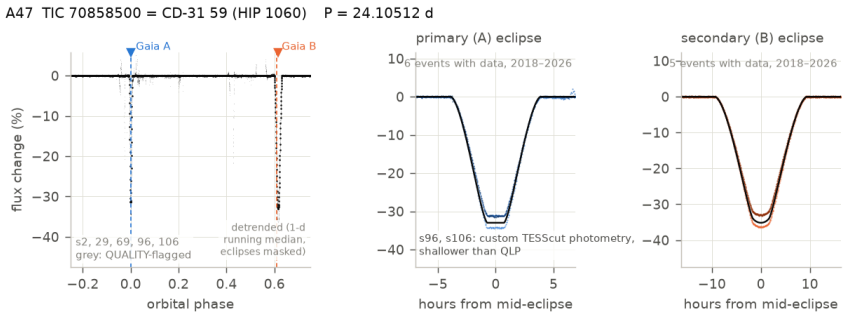


Figure 3. HIP 1060 (A47, TIC 70858500). Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.10512 d; dashed lines mark Gaia's predicted conjunctions A and B. Middle: the six 34%, 6.9-h total eclipses at A, 2018–2026. Right: the five 36%, 16-h annular transits at B.

Appointments. Of the 54 systems, 52 are long-period eclipsing binaries found at Gaia's conjunctions, with periods from 20.29 d (A35) to 57.24 d (A63). Two are by-catch at periods other than Gaia's: A45 (31.15 d, against a Gaia double-lined period of 21.68 d) and A54 (a 3.02-d eclipsing pair whose 39-d Gaia orbit is spurious). A third by-catch object, A46, is a 0.38-d contact or ellipsoidal variable on an unresolved pair and is not counted.

Period folds at P < 20 d. A5–A12 come from the folds at parallax > 10 mas and at G < 9; all eight were known spectroscopic binaries, and the eclipses are new. A5, A6, A7 and A10 are probable, because the fold makes period agreement certain by construction and none has a pixel test or a second pipeline. The fold at G 9–10 produced A75–A109, all 35 confirmed. A93 was already typed as an eclipsing binary in SIMBAD from a SuperWASP flag without a period (Schanche et al. 2019), so 34 are counted. In A95 the eclipse times vary with an amplitude of 3.4 ± 0.3 min on the 541-d period of Gaia's astrometric orbit for the star, consistent with the light-travel time that orbit predicts (chance coincidence ~10%).

M-dwarf searches. A1–A4 were detected as planet-like signals on nearby M or K dwarfs. A1, A2 and A4 are eclipses on a neighbouring star whose light enters the dwarf's aperture. A3 is on the target and is probable: Gaia shows that its companion is most likely a star, but the companion's orbital period is not known.

Edge-on astrometric orbits. A19 is probable: three eclipses repeat at a period that agrees with Gaia's to 0.7σ, but they fall about 2.5σ from the nearer predicted conjunction, and two exist only in flagged or QLP-only data. A72 eclipses on its 215.0116-d Gaia orbit itself. A74 is probable, with two 4.6% TESS eclipses three cycles apart and a ZTF night at the predicted time of the cycle between them. A73 (TYC 4590-364-1) is a 17.54-d eclipsing binary inside a 125-d Gaia orbit. Its eclipses have faded from 33% to 13% in four years and their timings vary by hours (Figure 4). A third body is needed to explain this, though the data do not show that it is Gaia's.

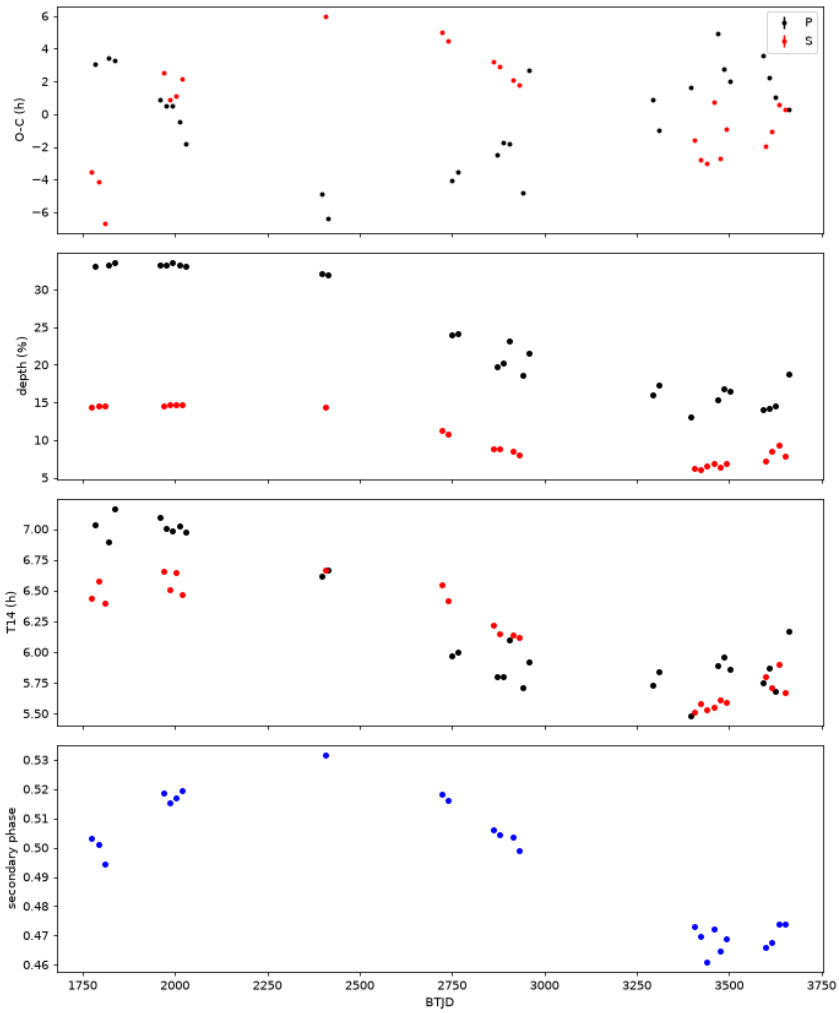


Figure 4. TYC 4590-364-1 (A73), from fits to individual primary (black) and secondary (red) eclipses in QLP sectors of 2019–2024. From top: timing residuals against a linear 17.54-d ephemeris, eclipse depths, total durations, and the phase of the secondary (blue). The eclipses fade from 33% to 13% and narrow, and the timings vary by hours.

Probable blends. Three signals from the crowded-field and known-host searches, A13–A15, are eclipses on faint stars near the targets, which the pixels locate but not to a single star or not with enough sectors. They are listed as probable and are not counted among the 104.

3.3 Heartbeat stars

Seven eccentric binaries show a brightening and dimming locked to periastron rather than an eclipse (Table 3). For each we fitted the equilibrium-tide model of Kumar et al. (1995) to the light curve alone and compared the e and ω it returns with Gaia's radial-velocity orbit. Three come from the appointments search. HD 193389 (Gaia P 20.7966 d) recovers Gaia's ω to 0.3° and e to 0.016 (3.2σ by Gaia's error); TYC 8203-301-1 also shows a coherent 842-ppm oscillation at 23.0009 ± 0.0006 times the orbital frequency, which we read as a tidally excited oscillation. Four more come from the G 9–10 fold. Of nine light curves set aside there for a brightening-and-dip feature, seven put it at Gaia's periastron (about 3 in 10,000 by chance); three of the seven are published heartbeat stars, recovered blind, and four are in no catalogue or heartbeat search. BD+61 2067, BD+37 4854 and CD-66 2078 are confirmed; TYC 9045-989-1 is a candidate, its ω 20° (2.4σ) from Gaia's.

3.4 Gaia DR3 orbit solutions contradicted by the eclipses

Eclipses fix the period, and their spacing fixes e cos ω, independently of any velocity measurement. We count a Gaia solution as contradicted when its period is spurious or wrong, or when the period or e cos ω differs from the eclipse value by 3σ or more of Gaia's quoted error, whether Gaia's covariance is propagated in full or its correlations are ignored, and however the ephemeris is carried. Conjunction offsets alone, and e and ω from eclipse durations, are recorded as tensions. Twelve of the 91 counted systems compared with a non-circular Gaia orbit meet the rule (Table 4): two periods are spurious or wrong (A54, A9), five are 3.4–5.0σ off, and five have e cos ω 3.7σ or more off. Three conjunction offsets (A89, A104, A87) move between about 2σ and about 8σ with the treatment and are not counted; A109's period difference falls to 1.8–2.7σ once its observed period change is included.

3.5 Catalogue corrections

Eleven catalogued stars have the wrong type, the wrong period or none (B1–B11, Table 5). Nine corrections are established by the data; two are proposed, B1 because its host is not yet separated from an unresolved neighbour 4″ away and B7 because a contamination ratio of 1.77 leaves a blend possible. V1274 Tau (B5) is typed RS at 6.62 d in the GCVS and 6.448 d in VSX, and ASAS-3 lists it as ESD/EC at 1.727 d (Pojmański 2002). TESS shows 4.5- and 2-ppt eclipses at Gaia's 5.62323 d, and the ASAS-3 photometry itself peaks at the 6.450-d rotation period with no power at 1.727 d. Five appointment systems (A20, A25, A31, A32, A52) are VSX type revisions.

3.6 Planet candidates and other objects

The M-dwarf search recovered both transit candidates of TIC 4206066 (Rabtsevich 2026) blind, at SNR 15 and 10 (C1; P = 11.132690 and 3.182785 d). TOI-5394.01 (HD 89277, C2) is the occultation of the M-dwarf companion in its Gaia SB1 orbit (P 15.195017 d), not a planet transit. The search around known hosts recovered the two TOI-6284 signals and a TOI-1467 threshold-crossing event, all in a preprint (C3; Tschudi 2026). Table 6 lists the other objects, for example GJ 4348, a young M dwarf that is plausibly a near-twin M+M pair at 0.223864 d, a case one spectrum at quadrature would settle.

Table 1. Yield of the appointments search.

	***G* < 10 subsample (1,422 stars)**	***G* 10–11 subsample (2,327 stars)**
Sent to independent re-analysis	16 stars (two more, HD 201073 and HD 7423, were dismissed on inspection before any check)	54 stars (TYC 5976-120-1, already in VSX as an eclipser, was not sent)
Eclipsing systems, uncatalogued	13 confirmed	35 confirmed + 4 probable (A37, A40, A53, A68) long-period; by-catch: TYC 447-247-1 (A54, 3.02 d) and TYC 7897-48-1 (A45, a 31.15-d eclipsing binary); BD+30 2277 (A46) is a contact or ellipsoidal variable and is not counted
Confirmed heartbeat stars, new	1 (HD 193389)	2 (TYC 8203-301-1, with tidally excited oscillations at 22, 23 and 24 times the orbital frequency; BD+59 2405)
Recovered as already known	20 catalogued eclipsing binaries recovered blind (not sent for re-analysis)	20 catalogued eclipsing binaries recovered blind; 3 on target (TOI-1897 and TOI-5149, planet candidates that are M-dwarf eclipsing binaries; TYC 1194-680-1, a SuperWASP EA with the wrong period); 2 known eclipsing binaries on neighbours 63″ away
Refuted	1 (systematics)	3 (all neighbour contamination at 36–60″; two of the neighbours are uncatalogued eclipsers themselves)
Plausible, unresolved	1 single event (BD+16 2639)	2 (TYC 7687-990-1, whose eclipses appear in 2025 but not 2019–2023; Wolf 73, a single event)

Table 2. Appointment candidates that are not new confirmed eclipsing binaries.

Object	Outcome	What the check found
Recovered, already known		
TOI-1897 = TIC 420645189	Known: retired planet candidate	TOI-1897.01 and .02 are this system's primary and secondary eclipses, retired by TFOPWG as FP/EB on 2024-09-17; P = 48.533998 ± 0.000022 d. Recovered blind. New: the Gaia SB1 gives f(M) = 0.0088 $M_\odot$, a 0.25 $M_\odot$ M-dwarf companion (K_1 12.4 km/s against about 50 m/s for the putative planet), and a sixth A eclipse in s86
TOI-5149.01 = TIC 286094277 = TYC 3578-1416-1	Known TOI; reinterpreted as an EBLM	A planet candidate since 2022-02-08; the 1.00–1.11%, 9.4–9.8-h transit is the TOI signal, P = 27.37163 ± 0.00006 d. New: a 602 ± 57 ppm secondary eclipse at phase 0.697 (z = 7.1; also in SPOC, TESS-SPOC and TARS); e cos ω = 0.3128 and e sin ω = −0.093 match Gaia; f(m) = 0.0014 gives m_2 = 0.11–0.13 $M_\odot$, R_2 = 0.15 $R_\odot$, T_2 ≈ 3100 K, so the companion is a low-mass star (an EBLM) rather than a planet. Speckle imaging shows a companion at 0.086″ (Δm 1.97)
TYC 1194-680-1 = TIC 435855781 = 1SWASP J004527.74+210427.9	Known eclipser; period and type revision	Listed as EA/EB with P = 1.66 d in Thiemann et al. 2021 (SuperWASP citizen science; not in VSX). TESS gives P = 22.450574 ± 0.000054 d (Gaia SB2 22.44398 ± 0.0033, 2.0σ off); 45% and 39% eclipses; joint SB2 fit: M = 1.144 + 1.161 $M_\odot$, R = 1.93 + 2.13 $R_\odot$, e = 0.1826, ω = 57.4°. This is a revision of the period and type, not a discovery
HD 181068 and seven catalogued EBs	Known; recovered blind	HD 181068, the Kepler triply eclipsing triple, recovered blind, and seven catalogued eclipsing binaries among the first 711 stars triaged: LW Pup, TIC 188625601, four IJspeert et al. 2024 OBAF eclipsers and one Prša et al. 2022 TESS EB. The full rerun of the G < 10 subsample recovered 20 known eclipsing binaries in all
Refuted		
TIC 71350905 = TYC 4361-615-1	Refuted: systematics	The claimed long periastron eclipse was orbit-edge scattered-light ramps: sector 40's first orbit, with no conjunction within 14 d, shows the same 1% drop. The mass function (0.0025 $M_\odot$) makes the companion a ~0.2 $M_\odot$ M dwarf whose transit could be at most 0.14% deep
TIC 212431508 = TYC 5162-1158-1	Refuted: neighbour, itself an uncatalogued eclipser	Both dips (13% and 4.3%) are on Gaia DR3 4230027841458799744 (G 11.02, 35.9″, same parallax, probably a wide companion), which loses 16.1% and 5.3% while the target loses 0.6% and −0.3%; depth falls 13 → 2.2 → 0.7% as the aperture tightens. The neighbour is itself a Gaia SB1 (P 33.294 d) whose A–B spacing is 2.9σ from its own orbit: a separate lead
TIC 17238861 = TYC 2508-895-1	Refuted: neighbour, itself an uncatalogued eclipser	The 1.9% dips are 10.5% and 9.6% V-shaped eclipses of TIC 17238865 = TYC 2508-1098-1 (G 12.20, 60.4″, same parallax but not bound); the target's core pixels change by no more than 0.1–0.4%. The neighbour's period is any of 732.0556/N d (N = 1–28); 48.80 d is one of these 28 aliases, the one tied to the target's orbit
TIC 265677843 = TYC 3335-389-1	Refuted: background eclipsing binary	The 7.37% dip is a 32.4% eclipse of Gaia DR3 250601205600534272 (G 12.06, 37″, parallax 0.85 mas, a background star); the target loses 0.01 ± 0.04%; TARS 0.83% against QLP 7.37%. Attribution to the target would also require a period 3.9σ from Gaia's
Plausible single events		
BD+16 2639 = TIC 334634917 = HIP 70251	Plausible, unconfirmed	One 0.52%, 2.50-h grazing eclipse at BTJD 2676.6639, 0.4σ from Gaia's A conjunction, identical in five pipelines and on the target's pixels, in the only TESS sector (s50). The geometry (i ≈ 86.3°) predicts the 2.5–3.0-h duration and no secondary. An unrelated dip has a chance of a few percent of falling inside a ±0.5 d window. Next A: BTJD 4345.49, 4365.60 (±0.17)

Object	Outcome	What the check found
Wolf 73 = HIP 6972 = TIC 365156806	Plausible, unconfirmed	One 2.98 ± 0.10%, 3.72-h flat-bottomed dip at BTJD 2481.9196 in SPOC 2-min data, 0.29σ from Gaia's B conjunction, on target; no other conjunction covered. A grazing occultation of a companion of at least 0.50 $M_\odot$. SPOC's own pipeline flagged the same event in 2021 at a spurious 32.139-d period. Next B: BTJD 4327.91 ± 0.47 (Gaia's period only)
Unresolved		
TYC 7687-990-1 = TIC 271053559	Plausible; eclipses only in 2025	In s89: a 31.33% V-shaped eclipse and two flat 5.19% and 5.51% eclipses, all on the target (difference images, TARS, clean common mode), at Gaia's 24.65-d period (P = 24.6548 ± 0.0018 d). However, the same ephemeris falls on covered, flat data in s8, s9 and s62, and a scan of every period from 24.58 to 24.76 d (the s89 value at ±50σ, Gaia's at ±5σ) finds, for each one, at least one of s8, s9, s35 and s62 covering the A conjunction with unflagged data and nothing deeper than 0.63% there, so the absence before 2025 is not an artefact of the one-sector ephemeris. Gaia also has a 300.8-d astrometric orbit on the star; our hypothesis, which is not proven, is a compact triple whose inner orbit precessed into eclipse between 2023 and 2025; the implied nodal-precession period of a few decades allows it. Next A: BTJD 4327.29, 4351.95 (±0.05; the third body can add hours)

Table 3. Heartbeat stars: six new and one candidate. The equilibrium-tide model of Kumar et al. (1995), fitted to the light curve alone, is compared with Gaia's independent orbit.

Name	TIC	T (mag)	P (d)	e	Origin	Status	Note
HD 193389	51223058	7.063	20.7974 ± 0.0015	0.36	App	new	F5.5 bright giant; the model recovers Gaia's ω to 0.3° and e to 0.016 (3.2σ)
BD+59 2405	405590521	9.886	25.90	0.59	App	new	A2 star in Trumpler 37; P is Gaia's
TYC 8203-301-1	147413927	10.134	22.0440 ± 0.0005	0.68	App	new	tidally excited oscillation: 842 ppm at 23.0009 ± 0.0006 times the orbital frequency, weaker peaks at 22 and 24 times
BD+61 2067	365670788	8.653	8.216284 ± 0.000094		Fold	new	the light curve alone returns Gaia's e and ω
BD+37 4854	333169278	9.38	5.077463 ± 0.000021		Fold	new	Gaia's ω returned once Doppler beaming is added
CD−66 2078	302055860	9.546	5.040480 ± 0.000030		Fold	new	Gaia SB2, near-twin F stars; returns Gaia's e and ω
TYC 9045-989-1	350856983	9.368	4.682234 ± 0.000021	0.12	Fold	candidate	Gaia SB2, near-twin F stars; ω 20° (2.4σ) from Gaia's; largely ellipsoidal

T: TESS magnitude from the TIC. e: Gaia's eccentricity where the text quotes it. HD 193389: Gaia SB1 P 20.7966 d. Origin as in Table A1: the first three were found by the appointments search (Section 2.5), the other four set aside by the G 9–10 Gaia-orbit fold (Section 2.4). None of the seven eclipses at the fitted inclination.

Table 4. Gaia DR3 orbit solutions contradicted by the eclipses: 12 of the 91 counted systems compared with a non-circular Gaia spectroscopic orbit. A solution is counted as contradicted when its period is spurious or wrong, or when its period or e cos ω differs from the eclipse value by 3σ or more of Gaia's quoted error under every treatment of its covariance.

Label	Name	Gaia P (d)	Eclipse P (d)	Fails in	Discrepancy
A9	HD 7601	13.42 (SB2)	27.0822013 ± 0.0000180	period (wrong)	$2P_{\rm Gaia}$ differs by 100σ; $(M_1 + M_2)\sin^3 i$ = 0.49 $M_\odot$, impossible for this pair
A12	HD 217492		14.0343984 ± 0.0000110	period	4.2σ (conjunction phase 5.5σ)
A21	TYC 4244-623-1		30.763077 ± 0.000013	e cos ω	3.7σ (RUWE 35)
A28	CD−32 4901	51.311 ± 0.014	51.37333 ± 0.00020	e cos ω; period	5.7–6.1σ; period 4.5σ
A29	BD−01 648	29.5888 ± 0.0053 (SB2)	29.60666 ± 0.00002	period	3.4σ
A30	BD+29 2845		27.94676 ± 0.00001	e cos ω	5.3–5.6σ
A52	TYC 5406-1905-1	40.543 ± 0.015 (SB2)	40.46953 ± 0.00005	period	4.9σ
A54	TYC 447-247-1	39 (SB2)	3.02348 ± 0.00001	period (spurious)	not an alias; would need 14 + 13.5 $M_\odot$
A63	TYC 3801-1499-1		57.2400 ± 0.0003	e cos ω	3.8σ (4.1–4.3σ from the eclipse spacing)
A65	BD+27 4098		26.768321 ± 0.000009	e cos ω	4.0–5.0σ in the eclipse spacing
A86	BD−01 2238		15.798442 ± 0.000003	period	4.1σ short (eclipse spacing 2.7σ)
A102	HD 194269		1.9385707 ± 0.0000005	period	5.0σ short

Discrepancies are in units of Gaia's quoted error. A blank Gaia period is not quoted in the text (for A30 and A65 it agrees with the eclipse period); the Supplement entries give it. Eclipse periods are those of Table A1, except for A29, where the value used in the comparison is given (Table A1: 29.606659 ± 0.000010 d). Three conjunction offsets are in tension without being counted, because they move with the treatment of Gaia's covariance: A87, 4.3σ at Gaia's epoch and 1.7σ at the TESS epoch carried with Gaia's period; A89, 6.7σ with the full covariance of a poorly fitting double-lined solution and 2.2σ with the correlations ignored; A104, 8.2σ with Gaia's epoch carried with the TESS period and 0.2σ with Gaia's own period (its eclipse spacing gives e cos ω 2.9σ from Gaia's).

Table 5. Catalogue corrections: the eleven numbered corrections B1–B11, the further corrections from the candidate checks, and the VSX type revisions among the new eclipsing systems.

Label	TIC	Name	Catalogue entry	TESS	Note
B1	48875187	Gaia DR3 5451130921097245056	VSX RS (Gaia classifier), no period	detached twin EB, P 0.862264 d, equal 9.1% eclipses	held: host not separated from two possible neighbours

Label	TIC	Name	Catalogue entry	TESS	Note
B2	382523691	ASASSN-V J005242.03-513706.0	VSX ROT 5.3637 d	EB at Gaia's SB1 period, *P* 4.915053 d, 24 + 12 ppt	
B3	397233697	Gaia DR3 522317746033943808	VSX ROT (Gaia); Gao ROT 8.40 d	EB, *P* 17.13557 d, 100 + 5 ppt	secondary at φ 0.3
B4	333792991	HAT-177-0002605	VSX VAR 9.69 d; ASAS-SN ROT:	EB, *P* 6.21467 d, 45 + 13 ppt	
B5	385557214	V1274 Tau	GCVS RS 6.62 d; VSX RS 6.448 d; ASAS-3 ESD/EC 1.727 d	EB at Gaia's *P* 5.62323 d, 4.5 + 2 ppt	ASAS-3 photometry shows only the 6.45-d rotation
B6	464315748	HD 171459	VSX DSCT|GDOR 3.19721 d	EB, *P* 6.3955 d, 12.7/13.4 ppt	3.19721 d is the eclipse spacing
B7	189269584	CD−28 7251	VSX VAR 0.0146 d	twin EB, *P* 6.9610 d, 20.7/23.9%	proposed: contamination ratio 1.77 leaves a blend possible
B8	99028319	LP 852-5	TESS-EB (Prša et al. 2022) P 9.4045943 ± 0.00045 d; not in VSX	EB, *P* 9.400436 d, 32.9/18.5 ppt	
B9	80158129	CzeV3749	VSX EA, no period	*P* 10.898518 d, no secondary	blend, 9″ from the peak
B10	120660110	WISE J105540.2-521127	VSX EW/EA 0.8152227 d	*P* 0.817052 d	blend, 11″
B11	306658481	NSV 3445	VSX suspected variable, no type or period	eclipsing, 1.66 d	1.5″; depth sector-dependent
Further corrections from the candidate checks					
	435855781	TYC 1194-680-1	SuperWASP EA/EB 1.66 d (Thiemann et al. 2021); not in VSX	EB, *P* 22.450574 ± 0.000054 d, 45% and 39%	Gaia SB2 22.44398 d, 2.0σ short
		Gaia DR3 3327090097954378752	VSX E 3.23146 d	EB, *P* 5.725588 ± 0.000018 d, circular	G 14.8, 63″ from TIC 206952704
		Gaia DR3 5956391969014516736	VSX E 12.3202 d	EB, *P* 1.5432 d	G 13.9, 53″ from TIC 34344890
VSX type revisions among the new eclipsing systems (Table A1)					
A20	235666667	HD 170829 = NSV 24480	VSX VAR:, no type or period	EB, *P* 26.384493 d	
A25	258780392	BD+12 121	VSX MISC 91.5 d	EB, *P* 43.8230 d	91.5 d is spot modulation of the giant
A31	2048935928	V1192 Cas	VSX SRA 37.4 d	EB, *P* 37.4133 d	37.4 d is the orbital period
A32	399758284	TYC 97-228-1	VSX ROT 39.81 d	EB, *P* 40.51032 d	39.81 d is the giant's near-synchronous rotation
A52	307170556	TYC 5406-1905-1	VSX MISC 79.3 d	EB, *P* 40.46953 d	

Depths are as measured in TESS (B1 in TGLC photometry). B1 and B7 are proposed and await the further step in their notes; the other nine are established by the data. For B9 and B10 the TIC star is the aperture in which the blended signal was found. The two Gaia DR3 stars of the second block are neighbours of conjunction-window targets, detected through their contamination of the targets' apertures. TYC 1194-680-1 is a confirmed eclipsing binary but not new: the revision is of its period and type. A21's only prior record is an ExoFOP CTOI that is the star's own secondary eclipse.

Table 6. Other objects: the planet candidates C1–C3, variables that are not eclipsing binaries, instructive false positives and rejected candidates, one line each; the Supplement has the details.

Object	TIC	What	Status
Planet candidates			
C1: TIC 4206066 b, c	4206066	two transiting-planet candidates (Rabtsevich 2026), *P* 3.182785 and 11.132690 d, recovered blind by the M-dwarf search	candidates; independently confirmed in public data; not our discovery
C2: TOI-5394.01 = HD 89277	61109252	the TOI signal is the occultation of an M-dwarf companion; *P* 15.195017 d is Gaia's SB1 period	explained
C3: TOI-6284, TOI-1467	286763141, 240968774	three transit signals around known hosts: 7.349277 and 5.244544 d (TOI-6284), 3.154127 d (TOI-1467)	independently recovered; in a preprint
Other objects			
GJ 4348	206327797	0.223864-d double wave; a detached near-twin M+M pair plausibly on top of a young star	open; one spectrum at quadrature settles it
TYC 3763-577-1	322347036	δ Scuti star, 15.32, 20.53 and 27.14 d^{-1}, ~1 ppt (diluted)	uncatalogued
TIC 234495456	234495456	BY Draconis spot rotator at ~0.7623 d	uncatalogued
TIC 189882802	189882802	a 3,444-ppm, 4-h single event made by two flares	dead
TIC 96202086	96202086	0.63048 d, 307 ± 32 ppm; ~40 such peaks expected over 1,299 stars	not a detection
HD 151770	337393383	real 16.3 + 11.2 ppt eclipses; $e \cos \omega$ ~7σ from Gaia's SB2 and both conjunctions miss	open
HD 34757, HD 44342, HD 196402	309416862, 38022998, 100614055	demoted fold candidates of 0.9–1.6 ppt	dead or unresolvable
EBLM J0021–16 = BD−16 52	406352710	published EB recovered blind by the *G* 9–10 fold; *P* 5.9672641 ± 0.0000012 d	known; a blind recovery
BD+30 9	22503559	ellipsoidal variable on a Gaia SB1 orbit, *P* 1.5786673 ± 0.0000010 d, 16.1 ppt peak to peak; no eclipse	new ellipsoidal variable
TYC 3707-251-1	316045876	4%, ~3-h dips every 5.854343 d; withdrawn as an eclipser on a 197-d Gaia astrometric orbit	new 5.854-d EB; triple probable, host not pixel-verified
TIC 271520202	271520202	two eclipses (18.5% and 17.3%) 4 × 192.391 d apart; Gaia 193.38 ± 0.81 d	plausible; one more primary would confirm
TIC 4382254	4382254	a lone 7,328-ppm, 0.72-h dip in sector 70	not a candidate
Marginal folds		eclipse-like folds in no catalogue	recorded, not chased

C3: all three signals are in the preprint of Tschudi (arXiv:2607.23781). TIC 316045876 and TIC 271520202 come from the edge-on astrometric sample (Section 2.6). The TIC numbers of the marginal folds and the Gaia binary suspects among bright late-M dwarfs are in the Supplement.

4. Discussion

4.1 Why the appointments method works, and its yield

Gaia's elements fix the conjunction times in advance: a single dip either lands in a window fixed beforehand or it does not. The search found 13 new eclipsing systems among 1,422 stars at G < 10 (0.9 per hundred) and 41, four of them probable, among 2,327 at G 10–11 (1.8 per hundred): 1.4 per hundred overall. The period fold of Gaia orbits at P < 20 d gave 0.4 per hundred at G < 9 and 1.3 at G 9–10. The searches were built differently and the rates are not directly comparable.

4.2 What the eclipses say about the Gaia orbits

Eclipses give the inclination and the relative radii that a spectroscopic orbit leaves free. Most systems with a Gaia orbit at the eclipse period agree with it within about 2σ. Twelve of the 91 counted systems compared with a non-circular Gaia orbit do not, by 3σ or more in period or $e\cos\omega$ under every treatment of Gaia's covariance (Section 3.4). Among the G 9–10 systems, 13 of 31 non-circular orbits have a largest offset of 2.3σ or more where about two are expected, and Gaia's quoted errors for these orbits appear to be underestimated by a factor of about 1.5.

In A54 a 3.02-d eclipsing pair sits under a spurious 39-d Gaia SB2 solution. El-Badry et al. (2026) found 14 hierarchical triples of the same kind, with the eclipse period much shorter than the Gaia period, and report about half of their Gaia SB1 solutions for compact-object candidates to be spurious. The eclipsers here are a different population, but the contradicted solutions show that a Gaia orbit requires independent confirmation.

4.3 The eclipses on the Harvard plates and in ASAS-SN

The ephemerides can also be run backwards. The Harvard plates, digitised by DASCH (Grindlay et al. 2012; data release 7), cover most of the sky from 1890 to 1990 with a scatter of about 0.13 mag per plate at G 10–11, enough to show a 10–40% eclipse if enough plates fall inside the window. The 31 systems among A1–A74 and B1–B11 with an eclipse of 8% or more and G ≤ 12.5 were folded back 600–12,000 cycles, with the cuts, the statistic and a permutation null fixed before the first fold. Seventeen of the 31 show the eclipse at our ephemeris, 15 of them at the TESS period unchanged; A30's 42% eclipse is a 0.6-mag dip on 27 plates (z = 19). Two systems are untestable, six are too shallow for their plates, and six are below threshold with depths consistent with TESS. The last of these, A63, was first read as an absence: nine plates in its window give 0.066 ± 0.036 mag against the 0.21 mag peak depth of its 18% eclipse, and injected eclipses were recovered 10 times in 10. But the injections were boxes at the peak depth and the eclipse is V-shaped, so the window mean it predicts is 0.105 mag, 1.1σ from the measurement; fitted with the TESS eclipse shape, the plates allow 0.62 ± 0.29 of the TESS depth, where nine of the detected systems fitted the same way give a median of 0.95 with a scatter of 0.25 between stars. A63 is consistent with its TESS depth, not absent. The same fold, run after submission on the five systems among A75–A109 that meet the cuts, adds A108 (TYC 2747-436-1), whose eclipse appears on 258 in-window plates from 1899 to 1990 (z = 19, the third-strongest detection); A83 is suggestive (FAP 0.005) at a period 4 eP below the TESS value; A82 and A86 are not seen, the plates allowing 0.04 ± 0.31 and 0.28 ± 0.23 of their TESS depths; A98 is untestable. Where TESS covers one or two sectors the century of baseline can refine the period: the plates detect A37 at 22.6131 ± 0.0002 d, twenty times tighter than the TESS value, which Table A1 keeps.

ASAS-SN V-band photometry (2014–2018; Shappee et al. 2014; Kochanek et al. 2017) gives a second archival test, independent of the plates and reaching shallower eclipses. Of the 118 objects with an ephemeris, 55 have an ASAS-SN light curve with a measured depth, a duration and an in-eclipse epoch. Folded on the ephemerides of Table A1, a backward extrapolation of 15 to 630 cycles into data that had no part in deriving them, and tested against a null that slides the eclipse window to 10,000 random phases, 20 of the 55 are fainter in eclipse at p < 0.05, where 2.75 are expected by chance, and six at p < 0.01 (A6, A33, A35, A38, A39, A47), where 0.55 are expected. Seven of the 20 are also on the plates (A4, A6, A33, A38, A47, A55, A56); ASAS-SN recovers 13 that the plate fold could not, four of which (A22, A50, A65, A70) the plates tested and did not detect. For the 11 systems detected under a shuffle-permutation null, measured and predicted depths agree (median ratio 1.04, Pearson r = 0.785, permutation p = 0.003).

ASAS-3 V photometry (2000–2009; Pojmański 2002) fills the gap between the plates and ASAS-SN for the 49 objects south of declination +28°, 34 of them with usable light curves. Folded on the same ephemerides with the same sliding-window null, nine are fainter in eclipse at p < 0.01 (A6, A10, A22, A28, A45, A47, A57, A58, A66), and A54 is a tenth once its window is narrowed to its 2-h eclipse; A47 measures 0.456 mag against 0.451 predicted, 193 cycles back. Five of the ten (A10, A45, A57, A58, A66) have no other non-TESS recovery, and A57's excess depth may be the giant's own variability. Among A75–A109, 13 of the 19 systems with usable curves are detected, out to 4,707 cycles (A106). Calibrating the sliding-window null on the light curves themselves, with 440,000 fake ephemerides placed away from the eclipses, gives a false-positive rate of 0.006 at p < 0.01 and 0.042 at p < 0.05, so these counts and the ASAS-SN counts are, if anything, conservative.

4.4 Limits

The completeness of the appointments search is not measured: no eclipses were injected into the 3,749 light curves, so the fraction of real eclipsers it finds and its false-alarm rate are not known. Seven of the 54 reported systems were admitted on the remaining pre-set cuts without the z test. The conjunction test inherits Gaia's period: A9 passed it at Gaia's 13.42 d, and its true period is 27.08 d. The literature check (Section 2.7) read the titles of the documents the NASA ADS full-text index returned, so a mention in the body or a table of an unrelated paper, or under a name not queried, could be missed; the novelty claim is conditional on that check. No spectroscopy was used. The sample stops at G 11.

4.5 Predicted eclipses and follow-up

Appendix A gives the next eclipses, in geocentric UTC, for every confirmed and probable system except A3. Most predictions are good to a few minutes. Most of the stars are G 9–11, with eclipses 1–40% deep lasting hours, so many are within reach of small telescopes. The TESS pipelines reduce the same pixels; an eclipse seen from the ground at the predicted time is the independent observation. A missed eclipse shows an ephemeris error, and an early or late one measures eclipse-timing variations (hinted at, for example, in A47). Timings of both eclipses measure $e \cos \omega$ and so test the contradicted Gaia orbits; depths of A73 would show whether its fade continues; timings of A95 would test the light-travel time. HIP 1060 (A47), with an annular eclipse on 2026 November 2 at 06:37 UTC, has been submitted to the AAVSO VSX and is pending.

The catalogue is bright (median T = 9.6). Every object saturates ZTF, the 27 with no ASAS-SN source are 0.81 mag brighter in the median than those with one, and ASAS-SN's scatter rises from 0.02 mag at V 11–12 to 0.08–0.09 mag brighter than V 9.5. For about a quarter of these systems a small telescope is the only instrument left that can follow them.

4.6 The null planet search

The transit searches of about 3,100 nearby M dwarfs (Section 2.3) found no new planet. Under the estimator's assumptions, the repeated-transit searches were expected to yield about 1.2 detections, with a probability of none of 0.30 and a range of about 1 to 8 allowed by those assumptions; the single-transit search, which the TOI list does not cover, expected about 3 more and is the only one with a real shortfall (2.5σ). A naive occurrence-rate calculation (Dressing and Charbonneau 2015) gives 75–95. The difference lies in the sample. Every search but the one around known hosts excluded stars with a TOI or CTOI; the cut removed 2.2% of the stars but all 41 known hosts of a detectable 0.5–15 d planet in this population (95% upper bound on the surviving fraction 7.3%). The searches recovered both TIC 4206066 candidates blind and showed TOI-5394.01 to be the occultation of an M-dwarf companion (Supplement, S3.7).

5. Conclusions

1. We report 104 eclipsing systems in public TESS photometry whose eclipses are in no vetted variable-star or eclipsing-binary catalogue: 93 confirmed and 11 probable.
2. Fifty-four came from eclipses by appointment. Among 3,749 stars at G < 11 it found about 1.4 new eclipsing systems per hundred stars; 50 are confirmed and four probable.
3. Forty-six came from transit searches of nearby M dwarfs and from period folds of Gaia orbits shorter than 20 d, 34 of them at G 9–10; four came from edge-on Gaia astrometric orbits.
4. The eclipses contradict twelve Gaia DR3 orbit solutions in period or $e \cos \omega$.
5. Six eccentric binaries are new heartbeat stars and a seventh is a candidate. We also report three probable blends and eleven catalogue corrections.
6. For 17 of 31 deep systems, Harvard plates from 1890–1990 show the eclipse at our ephemeris, and ASAS-SN photometry from 2014–2018 shows it for 20 of 55 systems tested, with the depths matching the predictions in the 11 examined.
7. A transit search of about 3,100 nearby M dwarfs found no new planet where about 1.2 detections were expected under the estimator's assumptions (probability of zero 0.30), because removing known planet candidates removed most stars where a detectable planet is known. Both TIC 4206066 candidates were recovered, and TOI-5394.01 is the occultation of an M-dwarf companion.

HIP 1060 (A47) has been submitted to the AAVSO VSX and is pending; submissions for the others are in preparation.

6. Contributions and AI use

The author posed the question, directed the work, commissioned and read an internal adversarial review of the draft, decided what is claimed and at what status, and takes responsibility for the content of this paper. The searches, measurements, checks and text were carried out by AI language-model agents running on the author's computer: Claude Opus 5.5, Claude Opus 5, Claude Fable 5 and Claude Fable 5.1 (Anthropic 2026), and GPT-5.6 and GPT-6 (OpenAI 2026). In outline, Claude Opus 5.5 designed and ran the M-dwarf transit searches and the first Gaia-orbit fold, explained TOI-5394.01, and carried out the independent re-analysis of every candidate of the appointments search, the G 9–10 fold and the edge-on search; Claude Opus 5 ran the late-M searches, the expected-yield analysis of Section 4.6, the live catalogue check and the ASAS-SN archival fold of Section 4.3; Claude Fable 5 ran the scaled Gaia-orbit fold and its continuation to G 9–10; Claude Fable 5.1 devised and ran the appointments

method, folded the Harvard plates and assembled the paper; GPT-5.6 ran the edge-on astrometric-orbit search and the look-elsewhere test for TIC 4206066; GPT-6 read and cross-checked the work and ran no searches. The per-object credit is recorded in the Supplement and in the deposit (Appendix B).

The adversarial review was run by a separate group of agents with its own code and no access to the agents writing the paper. It re-derived about 2,500 of the paper's numbers from the analysis files and re-reduced eight systems from the raw light curves; a second review of the same kind read the 35 systems added afterwards and re-measured eight of them. Both were internal quality control with independently written code, not journal peer review. Only public data were used; no telescope time, spectroscopy or proprietary data were involved. The use of AI systems is disclosed here and in the Acknowledgements, as the Journal's editorial policy asks.

7. Acknowledgements

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The analysis, the checks and the text of this paper were produced by AI language-model agents (Anthropic 2026; OpenAI 2026) directed by the author, as set out in Section 6. We thank P. Rabtsevich, whose public report of the TIC 4206066 candidates prompted the checks of Section 3.6.

References

Anthropic 2026, Claude Opus 5, Claude Opus 5.5, Claude Fable 5 and Claude Fable 5.1, language models used through Claude Code, https://www.anthropic.com/claude, accessed July–October 2026.

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, Astrophys. J., 935, 167 (DOI: 10.3847/1538-4357/ac7c74).

Bashi, D., Shahaf, S., Mazeh, T., et al. 2022, Mon. Not. Roy. Astron. Soc., 517, 3888 (DOI: 10.1093/mnras/stac2928).

Beck, P. G., Grossmann, D. H., Steinwender, L., et al. 2024, Astron. Astrophys., 682A, 7 (DOI: 10.1051/0004-6361/202346810).

Borkovits, T., Hajdu, T., Sztakovics, J., et al. 2016, Mon. Not. Roy. Astron. Soc., 455, 4136 (DOI: 10.1093/mnras/stv2530).

Bouma, L. G., Hartman, J. D., Bhatti, W., et al. 2019, Astrophys. J., Suppl. Ser., 245, 13 (DOI: 10.3847/1538-4365/ab4a7e) (CDIPS HLSP, DOI: 10.17909/t9-ayd0-k727).

Boyle, A. W., Bouma, L. G., and Mann, A. W. 2026, Astrophys. J., Suppl. Ser., 284, 75 (DOI: 10.3847/1538-4365/ae6657) (TARS HLSP, DOI: 10.17909/2sff-fn29).

Bryant, E. M., Bayliss, D., and Van Eylen, V. 2023, Mon. Not. Roy. Astron. Soc., 521, 3663 (DOI: 10.1093/mnras/stad626).

Caldwell, D. A., Tenenbaum, P., Twicken, J. D., et al. 2020, Res. Notes AAS, 4, 201 (DOI: 10.3847/2515-5172/abc9b3) (TESS-SPOC HLSP, DOI: 10.17909/t9-wpz1-8s54).

Callahan, J., Rowan, D. M., Kochanek, C. S., et al. 2025, Open J. Astrophys., 8, arXiv:2506.14869 (DOI: 10.33232/001c.142226).

Chen, X., Wang, S., Deng, L., et al. 2020, Astrophys. J., Suppl. Ser., 249, 18 (DOI: 10.3847/1538-4365/ab9cae).

Czavalinga, D. R., Mitnyan, T., Rappaport, S. A., et al. 2023a, Astron. Astrophys., 670A, 75 (DOI: 10.1051/0004-6361/202245300).

De Elías, J. 2025, J. Amer. Assoc. Var. Star Obs., 53, 3.

Drake, A. J., Graham, M. J., Djorgovski, S. G., et al. 2014, Astrophys. J., Suppl. Ser., 213, 9 (DOI: 10.1088/0067-0049/213/1/9).

Dressing, C. D., and Charbonneau, D. 2015, Astrophys. J., 807, 45 (DOI: 10.1088/0004-637X/807/1/45).

El-Badry, K., Latham, D. W., Rix, H.-W., et al. 2026, Open J. Astrophys., 9, arXiv:2608.06453 (DOI: 10.33232/001c.172274).

Eyer, L., Audard, M., Holl, B., et al. 2023, Astron. Astrophys., 674A, 13 (DOI: 10.1051/0004-6361/202244242).

Feinstein, A. D., Montet, B. T., Foreman-Mackey, D., et al. 2019, Publ. Astron. Soc. Pacific, 131, 094502 (DOI: 10.1088/1538-3873/ab291c) (eleanor HLSP, DOI: 10.17909/t9-ncv5-bb52).

Fuller, J. 2017, Mon. Not. Roy. Astron. Soc., 472, 1538 (DOI: 10.1093/mnras/stx2135).

Fuller, J., and Lai, D. 2012, Mon. Not. Roy. Astron. Soc., 420, 3126 (DOI: 10.1111/j.1365-2966.2011.20237.x).

Gaia Collaboration, Arenou, F., Babusiaux, C., et al. 2023, Astron. Astrophys., 674A, 34 (DOI: 10.1051/0004-6361/202243782).

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, Astron. Astrophys., 674A, 1 (DOI: 10.1051/0004-6361/202243940).

Gill, S., Cooke, B. F., Bayliss, D., et al. 2020, Mon. Not. Roy. Astron. Soc., 495, 2713 (DOI: 10.1093/mnras/staa1248).

Gosset, E., Damerdji, Y., Morel, T., et al. 2025, Astron. Astrophys., 693A, 124 (DOI: 10.1051/0004-6361/202450600).

Grindlay, J., Tang, S., Los, E., and Servillat, M. 2012, in New Horizons in Time-Domain Astronomy, IAU Symp. 285, 29 (DOI: 10.1017/S1743921312000166) (DASCH, https://dasch.cfa.harvard.edu, data release 7).

Guinan, E. F., Engle, S. G., and Devinney, E. J. 2012, J. Amer. Assoc. Var. Star Obs., 40, 467.

Halbwachs, J.-L., Pourbaix, D., Arenou, F., et al. 2023a, Astron. Astrophys., 674A, 9 (DOI: 10.1051/0004-6361/202243969).

Halbwachs, J.-L., Kiefer, F., and Lebreton, Y. 2023b, in SF2A-2023: Proc. Annual Meeting of the French Society of Astronomy and Astrophysics, 191.

Han, T., and Brandt, T. D. 2023, Astron. J., 165, 71 (DOI: 10.3847/1538-3881/acaaa7) (TGLC HLSP, DOI: 10.17909/610m-9474).

Handberg, R., Lund, M. N., White, T. R., et al. 2021, Astron. J., 162, 170 (DOI: 10.3847/1538-3881/ac09f1) (TASOC HLSP, DOI: 10.17909/t9-4smn-dx89).

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357 (DOI: 10.1038/s41586-020-2649-2).

Hartman, J. D., Bakos, G. Á., Bouma, L. G., et al. 2025, Publ. Astron. Soc. Pacific, 137, 024501 (DOI: 10.1088/1538-3873/adad42) (T16 HLSP, DOI: 10.17909/8nxx-tw70).

Hawthorn, F., Gill, S., Bayliss, D., et al. 2024, Mon. Not. Roy. Astron. Soc., 528, 1841 (DOI: 10.1093/mnras/stad3783).

Heinze, A. N., Tonry, J. L., Denneau, L., et al. 2018, Astron. J., 156, 241 (DOI: 10.3847/1538-3881/aae47f).

Huang, C. X., Vanderburg, A., Pál, A., et al. 2020, Res. Notes AAS, 4, 204 (DOI: 10.3847/2515-5172/abca2e) (QLP HLSP, DOI: 10.17909/t9-r086-e880).

Hunter, J. D. 2007, Comput. Sci. Eng., 9, 90 (DOI: 10.1109/MCSE.2007.55).

IJspeert, L. W., Tkachenko, A., Johnston, C., et al. 2021, Astron. Astrophys., 652A, 120 (DOI: 10.1051/0004-6361/202141489).

IJspeert, L. W., Tkachenko, A., Johnston, C., and Aerts, C. 2024, Astron. Astrophys., 691A, 242 (DOI: 10.1051/0004-6361/202450507).

Irwin, J. B. 1952, Astrophys. J., 116, 211 (DOI: 10.1086/145604).

Jayasinghe, T., Kochanek, C. S., Stanek, K. Z., et al. 2018, Mon. Not. Roy. Astron. Soc., 477, 3145 (DOI: 10.1093/mnras/sty838).

Jenkins, J. M., Twicken, J. D., McCauliff, S., et al. 2016, Proc. SPIE, 9913, 99133E (DOI: 10.1117/12.2233418).

Kirk, B., Conroy, K., Prša, A., et al. 2016, Astron. J., 151, 68 (DOI: 10.3847/0004-6256/151/3/68).

Kochanek, C. S., Shappee, B. J., Stanek, K. Z., et al. 2017, Publ. Astron. Soc. Pacific, 129, 104502 (DOI: 10.1088/1538-3873/aa80d9).

Kostov, V. B., Powell, B. P., Fornear, A. U., et al. 2025, Astrophys. J., Suppl. Ser., 279, 50 (DOI: 10.3847/1538-4365/ade2d8).

Kovács, G., Zucker, S., and Mazeh, T. 2002, Astron. Astrophys., 391, 369 (DOI: 10.1051/0004-6361:20020802).

Kumar, P., Ao, C. O., and Quataert, E. J. 1995, Astrophys. J., 449, 294 (DOI: 10.1086/176055).

Kunimoto, M., Huang, C., Tey, E., et al. 2021, Res. Notes AAS, 5, 234 (DOI: 10.3847/2515-5172/ac2ef0).

Laycock, S., Tang, S., Grindlay, J., et al. 2010, Astron. J., 140, 1062 (DOI: 10.1088/0004-6256/140/4/1062).

Li, M.-Y., Qian, S.-B., Zhou, A.-Y., et al. 2024a, Mon. Not. Roy. Astron. Soc., 534, 281 (DOI: 10.1093/mnras/stae2057).

Merle, T., Jorissen, A., Alexandre, S., et al. 2026, Mon. Not. Roy. Astron. Soc., 547, stag351 (DOI: 10.1093/mnras/stag351).

Mowlavi, N., Holl, B., Lecoeur-Taïbi, I., et al. 2023, Astron. Astrophys., 674A, 16 (DOI: 10.1051/0004-6361/202245330).

Ochsenbein, F., Bauer, P., and Marcout, J. 2000, Astron. Astrophys., Suppl. Ser., 143, 23 (DOI: 10.1051/aas:2000169).

Ogunwale, B. B., Zaguri, Y., Perdelwitz, V., et al. 2025, Astron. Astrophys., 704A, 317 (DOI: 10.1051/0004-6361/202555521) (TEQUILA HLSP, DOI: 10.17909/pj9v-7t21).

OpenAI 2026, GPT-5.6 and GPT-6, language models used through Codex, https://openai.com/, accessed July–October 2026.

Osborn, H. P., Armstrong, D. J., Brown, D. J. A., et al. 2016, Mon. Not. Roy. Astron. Soc., 457, 2273 (DOI: 10.1093/mnras/stw137).

Paegert, M., Stassun, K. G., Collins, K. A., et al. 2021, arXiv e-prints, arXiv:2108.04778 (https://arxiv.org/abs/2108.04778).

Pojmański, G. 2002, Acta Astron., 52, 397.

Powell, B. P., Kruse, E., Montet, B. T., et al. 2022, Res. Notes AAS, 6, 111 (DOI: 10.3847/2515-5172/ac74c4) (GSFC-ELEANOR-LITE HLSP, DOI: 10.17909/j2yt-t417).

Prša, A., Kochoska, A., Conroy, K. E., et al. 2022, Astrophys. J., Suppl. Ser., 258, 16 (DOI: 10.3847/1538-4365/ac324a).

Rabtsevich, P. 2026, report of two transiting-planet candidates around TIC 4206066: thread on X (https://x.com/p_rabtsevich/status/2107803696058822742) and write-up (https://tic4206066.pages.dev), both accessed 9 October 2026.

Rappaport, S., Deck, K., Levine, A., et al. 2013, Astrophys. J., 768, 33 (DOI: 10.1088/0004-637X/768/1/33).

Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2015, J. Astron. Telesc. Instrum. Syst., 1, 014003 (DOI: 10.1117/1.JATIS.1.1.014003).

Rimoldini, L., Holl, B., Gavras, P., et al. 2023, Astron. Astrophys., 674A, 14 (DOI: 10.1051/0004-6361/202245591).

Romanov, F. D. 2021, J. Amer. Assoc. Var. Star Obs., 49, 130 (https://arxiv.org/abs/2111.11086).

Rowan, D. M., Jayasinghe, T., Stanek, K. Z., et al. 2023, Mon. Not. Roy. Astron. Soc., 523, 2641 (DOI: 10.1093/mnras/stad1560).

Samolyk, G. 2022, J. Amer. Assoc. Var. Star Obs., 50, 255.

Samus', N. N., Kazarovets, E. V., Durlevich, O. V., et al. 2017, Astron. Rep., 61, 80 (DOI: 10.1134/S1063772917010085).

Schanche, N., Collier Cameron, A., Almenara, J. M., et al. 2019, Mon. Not. Roy. Astron. Soc., 488, 4905 (DOI: 10.1093/mnras/stz2064).

Schmidt, S. P., Schlaufman, K. C., Ding, K., et al. 2023, Astron. J., 166, 225 (DOI: 10.3847/1538-3881/ad0135).

Shappee, B. J., Prieto, J. L., Grupe, D., et al. 2014, Astrophys. J., 788, 48 (DOI: 10.1088/0004-637X/788/1/48).

Shporer, A., Fuller, J., Isaacson, H., et al. 2016, Astrophys. J., 829, 34 (DOI: 10.3847/0004-637X/829/1/34).

Solanki, S., Cieplak, A. M., Schnittman, J., et al. 2025, Astrophys. J., Suppl. Ser., 276, 17 (DOI: 10.3847/1538-4365/ad8a62).

Southworth, J. 2015, in Living Together: Planets, Host Stars and Binaries, ASP Conf. Ser. 496, Astron. Soc. Pacific, San Francisco, 164 (https://arxiv.org/abs/1411.1219).

Stassun, K. G., Oelkers, R. J., Paegert, M., et al. 2019, Astron. J., 158, 138 (DOI: 10.3847/1538-3881/ab3467).

Tang, S., Grindlay, J., Los, E., and Servillat, M. 2013, Publ. Astron. Soc. Pacific, 125, 857 (DOI: 10.1086/671760).

Tarrants, T., and Mendes, E. 2023, arXiv e-prints, arXiv:2310.12380 (https://arxiv.org/abs/2310.12380).

Thiemann, H. B., Norton, A. J., Dickinson, H. J., et al. 2021, Mon. Not. Roy. Astron. Soc., 502, 1299 (DOI: 10.1093/mnras/stab140).

Thompson, S. E., Everett, M., Mullally, F., et al. 2012, Astrophys. J., 753, 86 (DOI: 10.1088/0004-637X/753/1/86).

Triaud, A. H. M. J., Martin, D. V., Ségransan, D., et al. 2017, Astron. Astrophys., 608A, 129 (DOI: 10.1051/0004-6361/201730993).

Tschudi, Y. 2026, arXiv e-prints, arXiv:2607.23781 (https://arxiv.org/abs/2607.23781).

Watson, C. L., Henden, A. A., and Price, A. 2006, The Society for Astronomical Sciences 25th Annual Symposium on Telescope Science, The Society for Astronomical Sciences, Rancho Cucamonga, CA, 47.

Welsh, W. F., Orosz, J. A., Aerts, C., et al. 2011, Astrophys. J., Suppl. Ser., 197, 4 (DOI: 10.1088/0067-0049/197/1/4).

Wenger, M., Ochsenbein, F., Egret, D., et al. 2000, Astron. Astrophys., Suppl. Ser., 143, 9 (DOI: 10.1051/aas:2000332).

Williams, P. K. G. 2025, arXiv e-prints, arXiv:2501.12977 (https://arxiv.org/abs/2501.12977) (DASCH Data Release 7).

Yamaguchi, N., El-Badry, K., and Sorabella, N. 2024, Publ. Astron. Soc. Pacific, 136, 124202 (DOI: 10.1088/1538-3873/ad9955).

Zhang, H.-T., Qian, S.-B., Liao, W.-P., et al. 2026, Astrophys. J., 1008, 32 (DOI: 10.3847/1538-4357/ae8e79).

Appendix A. Ephemerides and next eclipses

Table A1 gives, for every confirmed and probable system, the period and epoch from the eclipses, the eclipse depths and the primary duration where measured, and the next predicted primary eclipse after the date of this paper (none for A3, which has no timing fit), in geocentric UTC (light-travel time to the barycentre removed). The next secondary eclipses are in the Supplement. Epochs are $\mathrm{BJD_{TDB}}$. The per-object entries in the Supplement give the measurements behind each row.

Table A1. Catalogue of the 104 eclipsing systems (93 confirmed, 11 probable) in label order, with the by-catch variable A46 and the previously typed A9 with unresolved hosts (A13–A15) in a separate block: ephemerides, depths and the next primary eclipse after 2026 October 14.

Label	TIC	Name	T (mag)	P (d)	T_0 ($\mathrm{BJD_{TDB}}$)	Depth prim. (%)	Depth sec. (%)	Dur. (h)
A1	311958594	UCAC4 083-054218	9.822	1.2443947 ± 0.0000016[a]	2458629.45849	2.28	1.85	0.7
A2	183494856	UCAC4 070-000968	14.810	6.129392 ± 0.000006	2457003.74608	21–29	17–23	
A3	85379511	HIP 83929 = G 181-29	9.656	9.474664[b]	2458993.379	0.0515		1.58
A4	70665829	UCAC4 537-050325	12.294	3.4657540 ± 0.0000093	2459528.2147 ± 0.002	30.2	6.73	2.75
A5	22388285	HD 106770	8.982	19.454722 ± 0.000010	2460494.45294 ± 0.00039	2.03		4.38
A6	348773952	BD−13 1046	9.596	5.0782783 ± 0.0000010	2460609.56921 ± 0.00016	10.97	2.79	2.55
A7	271321097	UCAC4 130-138342	10.930	13.9106635 ± 0.0000071	2460685.73790 ± 0.00034	8.00	1.00	3.8
A8	402800288	HD 131614	8.470	10.4462098 ± 0.0000098	2460391.45705 ± 0.00071	1.82		11.0
A9	399009076	HD 7601	8.557	27.0822013 ± 0.0000180[f]	2460097.60593 ± 0.00023	39.8	26.1	14.3
A10	300395201	HD 99010	8.449	7.8127421 ± 0.0000013	2460384.50567 ± 0.0001	2.20	1.73	4.0
A11	159892657	BD+40 3254	8.418	4.0751904 ± 0.0000009	2460133.82003 ± 0.00011	10.11	1.87	4.02
A12	436758044	HD 217492	8.326	14.0343984 ± 0.0000110	2460203.47040 ± 0.00028	1.82		4.0
A16	257352286	TYC 2173-965-1	9.423	23.802223 ± 0.000022	2459804.28519 ± 0.00089	10	8.45	20
A17	219230041	HD 26958	7.514	40.45701 ± 0.00002	2457386.08460	1.2		9.7
A18	189668511	TYC 2819-1027-1	9.216	31.80877 ± 0.00001[g]	2458766.58610	3.75	2.2	
A19	219766411	Gaia DR3 1440245818754478976	12.395	290.9037 ± 0.0006	2458936.3573 ± 0.0028	12.5		7
A20	235666667	HD 170829 = HR 6950 = NSV 24480	5.785	26.384493 ± 0.000040	2459403.97980	1.65	0.106	11.8
A21	403113935	TYC 4244-623-1	9.411	30.763077 ± 0.000013	2459849.89772 ± 0.00041	2.77	0.27	16.1
A22	324515186	HD 309050	9.444	23.034195 ± 0.000010	2459657.89250 ± 0.00031	8.3	1.1	7.5
A23	142935468	TYC 4160-760-1	9.454	22.244792 ± 0.000017	2459228.68805 ± 0.00056	1.70	1.8	6.4
A24	331507968	TYC 3669-262-1	9.278	24.690127 ± 0.000031	2458797.42249 ± 0.00146	0.86	0.32	6.9
A25	258780392	BD+12 121	8.520	43.8230 ± 0.0025	2459451.28700 ± 0.0013		2.9	16.5
A26	89031985	TYC 3091-707-1	8.791	42.14749 ± 0.00117	2459645.40345 ± 0.01221	0.29		18.8
A27	143198335	HD 63923	9.680	41.30385 ± 0.00030	2459963.04837 ± 0.00438		1.07	7.2
A28	145034339	CD−32 4901	9.251	51.37333 ± 0.00020	2458512.23071 ± 0.002	20–21	5	34.1
A29	248652521	BD−01 648	9.571	29.606659 ± 0.000010	2459177.83838 ± 0.0005	9.5		5.8
A30	311297982	BD+29 2845	9.595	27.94676 ± 0.00001	2458994.50441 ± 0.0018	42		10.1
A31	2048935928	V1192 Cas (= TIC 271237873)	9.791	37.4133 ± 0.0005	2458764.90610 ± 0.018	4.7	0.9	29
A32	399758284	TYC 97-228-1	9.956	40.51032 ± 0.00011[g]	2458463.92360 ± 0.024	1.3–1.4	0.45	
A33	4341778	TYC 7691-3893-1	9.825	45.59795 ± 0.00008[g]	2458573.67953 ± 0.0042	9	2.2	
A34	139285862	CD−45 14957	9.926	45.161458 ± 0.000109	2458354.30334 ± 0.00477	2.3		16
A35	11488700	CD−29 366 (Bok I 16B)	10.299	20.289728 ± 0.000160	2458383.84197 ± 0.00342	3.3		5.9
A36	43473641	TYC 5326-1527-1	10.344	54.851934 ± 0.000191	2459192.78258 ± 0.0053	1.0	0.39	29
A37	24755724	CPD−25 5943	10.213	22.611537 ± 0.002850	2460793.29985 ± 0.002	18.7	13.4	17.5
A38	316326405	TYC 3715-870-1	9.933	37.116502 ± 0.000308[g]	2458839.85111 ± 0.00926	8.7	2.5	
A39	22620802	TYC 3396-1373-1	10.210	35.692330 ± 0.000062	2459602.95779 ± 0.0007	2.6	1.25	8.7
A40	165541068	TYC 986-148-1	10.156	24.7458 ± 0.0025[c]	2460463.59430 ± 0.001	4.6	1.5	13.7
A41	156182608	TYC 6568-1036-1	10.521	23.53654 ± 0.00017	2458501.94000 ± 0.011		0.7–1.0	3
A42	832117928	CD−25 5638B	10.501	22.029385 ± 0.000018	2458508.34974 ± 0.00098	0.5		4.3
A43	241445833	TYC 4685-2083-1	10.376	23.205492 ± 0.000007	2458389.70430 ± 0.00052	6.9		4.25
A44	355477435	TYC 8461-136-1	10.501	21.106026 ± 0.000006	2458325.49469 ± 0.00051	5.7		3.1
A45	34344890	TYC 7897-48-1	10.495	31.152794 ± 0.000072	2459382.75960 ± 0.0037	6–7		3.8
A46	383529972	BD+30 2277 = WDS ES 436	10.118	0.381845 ± 0.000002	2459645.01110 ± 0.0007	4.6	3.4	
A47	70858500	CD−31 59 = HIP 1060	10.005	24.105122 ± 0.000006[g]	2458357.74508 ± 0.00079	36	34	
A48	66900858	TYC 3166-453-1	10.533	43.196065 ± 0.000034	2459804.38481 ± 0.00045	27	10.4	9.7
A49	286926100	TYC 3870-1689-1	10.010	28.098880 ± 0.000013	2459645.01560 ± 0.00026	2.6	0.45	5.8
A50	450218810	TYC 3729-162-1	10.396	24.172470 ± 0.000015	2459926.66270 ± 0.00041	11.5	1.6	7.1
A51	18000700	BD+28 4244	10.192	29.32848 ± 0.00033	2460558.15104 ± 0.060		1.1–1.3	16.4
A52	307170556	TYC 5406-1905-1	9.821	40.46953 ± 0.00005	2459206.58750 ± 0.0019	6		16
A53	96844936	TYC 8355-1052-1	10.463	36.7755 ± 0.006[d]	2461155.84550 ± 0.0007		1.8–2.4	5.9
A54	108344775	TYC 447-247-1	10.291	3.02348 ± 0.00001	2460482.03790 ± 0.0001	18.8	11.7	2.70
A55	11156440	BD+60 397	9.529	21.254843 ± 0.000012	2458801.37455 ± 0.00065	19.5	18.1	11
A56	367706305	HD 279736	10.354	32.286594 ± 0.000012	2458817.15278 ± 0.00047	17.5	17.2	14
A57	145469097	TYC 7121-1393-1	10.021	43.324287 ± 0.000075[g]	2458530.31198 ± 0.00384	5.8	2.2	
A58	421900559	BD−14 1559	9.563	35.976045 ± 0.000018	2459203.38951	9.8		6
A59	372183259	TYC 3187-1132-1	10.394	42.648930 ± 0.000055	2460537.57169	12.3		4
A60	142084127	TYC 9189-1130-1	10.022	45.49880 ± 0.00004[g]	2459553.26450 ± 0.00083	0.8	0.3	
A61	468529113	TYC 4308-545-1	9.981	21.552334 ± 0.000007	2459481.89522 ± 0.00022	14.6		4.9
A62	464830297	TYC 110-583-1	10.350	22.34089 ± 0.00001[g]	2458440.02660 ± 0.0004	2.1	0.15	
A63	467158573	TYC 3801-1499-1	10.500	57.2400 ± 0.0003	2458868.88650 ± 0.005	18	16	13
A64	249895319	BD+50 578	9.770	29.27232 ± 0.00005	2458793.94630 ± 0.0007	1.67	0.15	9.9

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h)
A65	88619181	BD+27 4098	10.018	26.768321 ± 0.000009	2459819.55960 ± 0.00015	10	0.72	7.2
A66	260563222	CD−41 5677	9.652	38.791921 ± 0.000024	2458553.49853 ± 0.00089	3.0	0.22	14
A67	173468180	TYC 3623-443-1	10.296	29.182072 ± 0.000005[g]	2458756.76116 ± 0.00042	3.8	2.3	
A68	80107967	TYC 8136-2645-1	10.580	34.184472 ± 0.000076	2458521.30741 ± 0.00302	1.3		11.4
A69	377060838	TYC 243-1377-1	10.162	42.463964 ± 0.000016[g]	2458517.53332 ± 0.00059	17	14.6	
A70	70789999	TYC 4094-1571-1	9.761	22.735028 ± 0.000008	2458823.53837 ± 0.00039	8.2	5.1	13
A71	87303050	TYC 3747-2187-1	10.200	24.445338 ± 0.000011	2458833.11980 ± 0.00047	9.7	1.4	4.5
A72	279636729	Gaia DR3 4720373709202359808	12.008	215.0116 ± 0.0002	2458341.4321 ± 0.0009	38	24.2	10.6
A73	315721826	TYC 4590-364-1	10.808	17.538[e]	2459940.800	33 to 13	14.6 to 6.3	7.0 to 5.6
A74	427345302		12.10	173.8755 ± 0.0005	2459884.9860 ± 0.0013	4.62	~5 (ZTF g)	10.8
A75	202282515	TYC 3255-1824-1	9.655	6.929866 ± 0.000015	2460111.42326 ± 0.00065	6.76	1.55	5.00
A76	202710940	TYC 3256-1533-1	9.480	3.3063718 ± 0.0000018	2460299.60466 ± 0.00020	0.81	0.78	2.25
A77	196475343	BD+44 219	9.116	6.370415 ± 0.000005	2460407.33044 ± 0.00027	0.6	0.37	1.9
A78	268587594	HD 6217	8.837	9.140474 ± 0.000004	2458764.02178 ± 0.00018	3.65	<0.4	2.38
A79	416277351	BD+64 520A	9.362	6.607651 ± 0.000012	2460149.97660 ± 0.00045	0.454	0.169	2.54
A80	19702288	TYC 1310-1984-1	9.479	17.02161 ± 0.00032	2459516.54908 ± 0.00042	7.98	0.77	7.4
A81	109061795	CD−24 5530	9.338	13.2271715 ± 0.0000023	2460040.42781 ± 0.00013	2.56	0.189	7.25
A82	317250623	CD−83 126	8.980	19.605278 ± 0.000014	2460060.10199 ± 0.00051	9.8	2.06	5.65
A83	7060775	BD+13 1919	9.163	6.422180 ± 0.000001	2459741.55165 ± 0.00012	10.1	4.59	4.15
A84	456339196	TYC 3805-1342-1	9.566	7.1061147 ± 0.0000043	2459530.89350 ± 0.00012	7.86	4.13	8.60
A85	38388435	CD−27 6336	9.141	2.6164745 ± 0.0000005	2460518.67725 ± 0.00008	5.5	0.28	3.6
A86	170891257	BD−01 2238	9.414	15.798442 ± 0.000003	2459811.62273 ± 0.00024	12	1.4	3.5
A87	463896609	HD 89725	8.885	2.2366733 ± 0.0000008	2460195.87831 ± 0.00025	1.64	0.36	4.3
A88	302351416	TYC 3827-170-1	9.422	13.9574485 ± 0.0000066	2459928.45698 ± 0.00023	3.57	<0.4	4.50
A89	388287047	CPD−85 289	9.317	17.638544 ± 0.000019	2460032.65988 ± 0.00069	1.61	<0.2	3.9
A90	454747287	CD−73 685	9.346	5.6347635 ± 0.0000015	2459691.10727 ± 0.00013	1.17	0.049	4.30
A91	311635507	HD 311112	9.375	18.5428744 ± 0.0000049	2460479.18152 ± 0.00017	6.67	0.87	7.27
A92	148877735	BD+69 653	9.325	3.6074499 ± 0.0000008	2460110.82736 ± 0.00008	1.32	1.32	3.05
A93	175734936	HD 112601	8.955	4.337953 ± 0.000001	2460380.72231 ± 0.00018	0.99	0.93	2.55
A94	245194662	HD 114997	8.917	17.2795467 ± 0.0000052	2460353.15051 ± 0.00022	2.8	0.25	4.2
A95	154346187	HD 234179	8.984	3.549541 ± 0.000002	2459489.65835 ± 0.00031	0.559	0.509	2.20
A96	426877400	HD 140021	9.623	2.7560089 ± 0.0000006	2460574.15160 ± 0.00014	0.76	0.031	4.5
A97	351595900	TYC 1529-387-1	9.316	6.647424 ± 0.000005	2459826.14200 ± 0.00052	0.59	<0.2	4.1
A98	157469799	BD+45 2574 A	8.948	3.5063244 ± 0.0000002	2460172.71372 ± 0.00002	24.5	19.6	3.85
A99	352236055	TYC 3925-1852-1	9.283	18.821610 ± 0.000008	2460160.83034 ± 0.00023	1.49	<0.13	4.7
A100	425304484	HD 353655	9.464	16.260285 ± 0.000024	2460531.06071 ± 0.00039	3.6	1.3	12.3
A101	14596588	HD 192721	8.889	4.478718 ± 0.000065	2460818.15412 ± 0.00013	1.44	0.277	2.95
A102	72160748	HD 194269	8.652	1.9385707 ± 0.0000005	2460815.72949 ± 0.00012	6.95	0.71	2.3
A103	188258837	BD+41 3794	8.916	5.9855010 ± 0.0000159	2460144.94717 ± 0.00108	3.62	3.18	11.0
A104	353453805	BD+42 3880	9.503	14.3509487 ± 0.0000051	2460309.02608 ± 0.00010	6.09	1.46	8.60
A105	321718006	BD+67 1313	8.837	3.9253291 ± 0.0000007	2460144.07847 ± 0.00008	1.18	0.09	5.6
A106	352935659	HD 205273	8.707	1.3854241 ± 0.0000004	2460402.27646 ± 0.00009	3.06	2.92	1.95
A107	397680054	BD+32 4452	9.490	2.976903 ± 0.000001	2459681.70466 ± 0.00012	5.57	5.52	3.2
A108	239649487	TYC 2747-436-1	9.131	6.4161826 ± 0.0000024	2460233.51625 ± 0.00014	21.4	2.8	12.6
A109	274331697	BD+43 1585	9.616	1.0394878 ± 0.0000005	2460134.11958 ± 0.00009	1.26	0.15	1.3
Probable eclipsing binaries with unresolved hosts (not counted)								
A13	58553437		10.228	1.078685		0.126		
A14	71124276		10.504	1.123892		0.082	0.022	
A15	261135533	TOI-6255	10.446	14.486682		0.1399		

Note: T: TESS magnitude from the TIC (every row). P and T_0: the ephemeris of the independent re-analysis or timing fit, to its quoted precision; T_0 is the epoch of the primary (deeper) eclipse, $\mathrm{BJD_{TDB}}$ = BTJD + 2457000. Depths and durations are aperture values in the named pipeline's flux (Section 2.8), not corrected for dilution. For A25, A27, A41 and A53 one eclipse kind has been observed; the source tabulates its depth in the secondary column, and its times are those given as the next primary. A51's primary is seen only as an ingress of at least 8.6%. Next primary: geocentric UTC, from $\mathrm{BJD_{TDB}}$ with the light-travel time to the solar-system barycentre in the star's direction removed (up to ±8.3 min) and TT − UTC = 69.184 s; an observer's local correction is at most 21 ms. Most predictions are good to a few minutes; A31, A32, A35–A38, A40 and A41 to 0.02–0.10 d; A1 and A2 to ±0.004 d and A19 to ±0.005 d (formal errors of the discovery ephemerides); A72 to ±0.002 d; A73 to ±0.3 d. None of these includes event-to-event timing scatter or A109's eclipse-timing variations of about 30 min. The Supplement entries give the secondary epochs and the next secondaries. [a] A1: T_0 is the shallower eclipse; the next primary listed is the deeper one. [b] A3: fold period and box epoch; no timing fit, so no prediction. [c] A40: Gaia's period (one event of each kind). [d] A53: the single interval between its two events. [e] A73: mean period of the primaries, which varies (single cycles 17.39–17.85 d); T_0 is the first of two clean, consecutive primaries (BTJD 2940.800 and 2958.651); the eclipses are fading, and duration and depths give the range (Section 3.2). [f] A9: the period first catalogued, 13.4166 d, was a fold alias. [g] The deeper eclipse is at conjunction B of Gaia's orbit; T_0, the depths and the next primary refer to it, and its duration, which differs from the A eclipse's, is in the Supplement. Origin: App, eclipses by appointment (Section 2.5); Fold, Gaia-orbit period fold at $P < 20$ d (Section 2.4); MD, M-dwarf transit searches (Section 2.3); Ast, edge-on Gaia astrometric orbit (Section 2.6). Notes: SB, a known spectroscopic binary before this work (the eclipses are new); G, Gaia DR3 orbit solution contradicted by the eclipses (Table 4); K, in the unvetted machine-learning list of Kostov et al. (2025, table 2); TCE, an unvetted SPOC or TESS-SPOC threshold-crossing event (72 of the 104); V, a VSX entry of a non-eclipsing type (Table 5); CTOI, an ExoFOP community TOI (its own secondary eclipse); R, a prior mention of the eclipses (Bryant et al. 2023); HP, eclipse detected on the Harvard plates (17 of the 31 examined); H, host component or a neighbour within one TESS pixel not separated; bc, by-catch, found at a period other than that of the Gaia orbit.

Appendix B. Data and code availability

All input data are public: TESS light curves, target-pixel files and full-frame-image cutouts from MAST; Gaia DR3 from the ESA Gaia archive; catalogues from VizieR, SIMBAD, ExoFOP, the NASA Exoplanet Archive and the AAVSO VSX; the Harvard plates from DASCH. The analysis code (Python, with NumPy, Astropy, Matplotlib and Requests), the per-object re-analysis reports, the fitted ephemerides behind Table A1, the figure sources, the object table (`tables/objects.csv`, one row per object with the analysis file behind it) and the Supplement are in a public data deposit, available from the author, under CC BY 4.0, with the code also under the MIT licence. The code was written for this work and has not been packaged for reuse; every light curve it reads is re-fetched from MAST. Files in the deposit written before 2026 October 12 use an older labelling in which this paper's A20–A71 are A21–A72 (HIP 1060, A47 here, is their A48); the object table uses this paper's labels throughout. The ASAS-SN archival fold of Section 4.3 (2026 October 14) postdates the deposit version cited above and is added in its next version. The version of this paper as submitted to the journal, whose main text carries the full entries for the catalogue corrections B1–B11, the probable blends A13–A15 and the other objects of Table 6, is in the deposit and in the public repository (github.com/Snowstar38/eclipses-by-appointment). The report of the internal adversarial review, the response to it and the consistency checks are kept with the paper's sources in the same deposit.